

5-Stage CMOS Ring Oscillator Design and Analysis using LTspice (TSMC 180 nm)

1. Objective

The objective of this project was to design and simulate a **5-stage CMOS Ring Oscillator** using **TSMC 180 nm BSIM3 (Level-49) transistor models** in LTspice. The design aims to generate a stable oscillation without requiring an external clock source while analysing its frequency, propagation delay, and startup behaviour.

2. Design Specifications

Parameter	Value
Technology	TSMC 180 nm CMOS
Simulator	LTspice XVII
CMOS Model	BSIM3 Level-49
Supply Voltage	1.8 V
Number of Inverters	5
Oscillator Type	Odd-stage CMOS Ring Oscillator
PMOS Length	180 nm
PMOS Width	2 μm
NMOS Length	180 nm
NMOS Width	1 μm
Output Load	1 fF
Simulation Time	100 ns

3. Theory

A ring oscillator consists of an **odd number of CMOS inverters** connected in a closed loop.

Each inverter contributes:

- One inversion (180° phase shift)
- One propagation delay

For oscillation,

- Total phase shift = 360°
- Loop gain ≥ 1

Since five inverters produce:

$$5 \times 180^\circ = 900^\circ$$

which is equivalent to

$$900^\circ = 720^\circ + 180^\circ$$

and the propagation delay contributes the remaining phase shift, sustained oscillation occurs.

The oscillation frequency is

$$f = \frac{1}{2Nt_p}$$

where

N = Number of inverter stages

t_p = propagation delay of one stage

4. Circuit Implementation

The oscillator consists of

- Five CMOS inverters
- PMOS connected to VDD
- NMOS connected to Ground
- Output of fifth inverter connected back to first inverter input

No external clock source is used.

Oscillation starts due to the inherent propagation delay of the inverter chain.

5. Challenges Encountered

Several issues were encountered during the design process.

Initial Problems

- Incorrect model library inclusion
 - Unsupported BSIM4 model
 - Wrong transistor model names
 - LTspice file path issues
 - Startup convergence problems
 - Perfect symmetry causing delayed startup
 - Incorrect transient analysis settings
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6. Modifications Performed

The following improvements were implemented.

(a) Technology Model

Initially,

Level-1 MOS models

were used.

These were replaced with

TSMC180nm BSIM3 Level-49

models for industry-level accuracy.

(b) Correct Model Names

NMOS

TSMC180nmN

PMOS

TSMC180nmP

(c) Transistor Dimensions

Final dimensions selected:

NMOS

$L = 180 \text{ nm}$

$W = 1 \text{ }\mu\text{m}$

PMOS

$L = 180 \text{ nm}$

$W = 2 \text{ }\mu\text{m}$

Reason:

PMOS mobility is approximately half of NMOS mobility.

Doubling PMOS width balances pull-up and pull-down strengths.

(d) Supply Voltage

$V_{DD} = 1.8 \text{ V}$

This matches the nominal operating voltage of TSMC 180 nm CMOS technology.

(e) Simulation Command

Final transient simulation

.tran 0 100n

(f) Output Capacitor

A small

1 fF

capacitive load was connected to the output node to represent parasitic loading.

7. Final Waveform Analysis

The final simulation shows

- Stable oscillation
- Continuous oscillation
- Rail-to-rail output swing
- Proper phase shift between adjacent stages
- Uniform oscillation frequency

The startup delay is approximately

2 ns

which is expected because oscillation begins only after the signal propagates around the loop.

8. Measured Results

Oscillation Period

Measured using

`.meas`

Result

$$T = 6.097 \times 10^{-10} \text{ s}$$

or

$$T = 0.6097 \text{ ns}$$

Oscillation Frequency

Measured

$$f = \frac{1}{T}$$

Result

$$f = 1.640 \text{ GHz}$$

Propagation Delay

Measured

$$t_p = 57.1 \text{ ps}$$

9. Theoretical Verification

Theory

$$f = \frac{1}{2Nt_p}$$

Given

N=5

tp=57.1 ps

Therefore

$$f = \frac{1}{2(5)(57.1 \times 10^{-12})}$$
$$f = 1.75 \text{ GHz}$$

Measured

$$1.64 \text{ GHz}$$

Difference

≈6%

This difference is expected because the theoretical equation ignores

- Junction capacitances
- Wiring parasitics
- BSIM nonlinear effects
- Output loading
- Finite rise/fall times

Hence, the measured result closely agrees with theory.

10. Advantages

- Extremely simple architecture
- No external clock required
- Fully digital implementation
- Small silicon area
- High oscillation frequency

- Easily scalable
 - Used in modern ASICs and SoCs
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11. Applications

- PLLs
 - Clock generation
 - Process monitors
 - Temperature sensors
 - Voltage Controlled Oscillators
 - Random Number Generators
 - Built-In Self-Test (BIST)
 - Time-to-Digital Converters
 - PUFs (Physical Unclonable Functions)
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12. Limitations

- Frequency varies with temperature
- Frequency varies with supply voltage
- Process variation affects delay
- Higher phase noise than crystal oscillators
- Sensitive to transistor mismatch